

Hevo Assessment 1

2. README.md

Hevo Assessment I: Data Pipeline Setup & Transformation

2.1. Project Overview

- **Goal:** Set up a CDC-based data pipeline from a local PostgreSQL source to a Snowflake destination using Hevo Data, including data transformation.

2.2. Prerequisites & Accounts Used

- **Hevo Account Details:**
 - **Team Name:** siddharthkakade1999@gmail.com
 - **Pipeline ID:** #7
- **Local Postgres Setup:** Docker desktop running latest postgres image

2.3. Detailed Steps to Reproduce the Setup

2.3.1. Database Setup (PostgreSQL)

1. Installation:

Command ran in docker desktop:

```
docker run --name mydatabase2 -e POSTGRES_PASSWORD=admin -e POSTGRES_USER=admin -e POSTGRES_DB=mydatabase2 -p 5433:5432 -d postgres
```

2. Schema Creation in pg4admin:

- a. Customer table:

customer

×

General
Columns
Advanced
Constraints
Partitions
Parameters
Security
SQL

Inherited from table(s)

Select to inherit from...

Columns

+

	Name	Data type	Length/Precision	Scale	Not NULL?	Primary key?	Default
	id	integer			☐	☒	
	first_name	character varying			☐	☐	
	last_name	character varying			☐	☐	
	email	character varying			☐	☐	
	address	json			☐	☐	

b. Feedback table:

Create - Table

×

General
Columns
Advanced
Constraints
Partitions
Parameters
Security
SQL

Inherited from table(s)

Select to inherit from...

Columns

+

	Name	Data type	Length/Precision	Scale	Not NULL?	Primary key?	Default
	id	integer			☐	☒	
	order_id	integer			☐	☐	
	feedback_comment	text			☐	☐	
	rating	integer			☐	☐	

?

?

×

 Close

↺

 Reset

💾

 Save

Create - Table

×

General
Columns
Advanced
Constraints
Partitions
Parameters
Security
SQL

Primary Key
Foreign Key
Check
Unique
Exclude

Name	Columns	Referenced Table
order_id	(order_id) -> (id)	public.orders

General

Definition

Columns

Action

Columns

+

Local column	Select an item...
References	Select an item...
Referencing	Select an item...

Add

Local	Referenced	Referenced Table
order_id	id	public.orders

?

?

×

 Close

↺

 Reset

💾

 Save

c. Order table:

	Name	Data type	Length/Precision	Scale	Not NULL?	Primary key?	Default
	id	integer			☒	☒	
	status	text[]			☒	☐	
	customer_id	integer			☒	☐	

Name	Columns	Referenced Table
customer_id	(customer_id) -> (id)	public.customer

Local	Referenced	Referenced Table
customer_id	id	public.customer

3. Data Loading: pg4admin along with CSV available on [GitHub - muskan-kesharwani-hevo/hevo-assessment-csv](https://github.com/muskan-kesharwani-hevo/hevo-assessment-csv): CSV files for Hevo interview assessment

a. Further CSV was imported by following -
<https://www.youtube.com/watch?v=lkd2xSb00UI>

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4. Networking/Connectivity:

- Command to enter the docker container to run the ngrok util: `docker exec -it mydatabase /bin/bash`
- Using ngrok command: `ngrok tcp 5432`
- Further `wal_level` was set to 10
- Restart agent service and granted the permission along with whitelisting the URL

2.3.2. Hevo Pipeline Setup (Objective 5)

1. When you open a Snowflake account, you will see a window that gives you the option to ingest data using Hevo pipelines. Selecting this option will take you to the Hevo User Interface (UI). In the Hevo UI, you need to configure the following settings for pipelines:
 - a. Choose the source type from which you want to ingest data; Hevo supports a variety of source types including databases, applications, and flat files.
 - b. Configure the connection details for your chosen source, such as connection strings, container names, or file paths.
 - c. Set up the destination where the data will be loaded, in this case, your Snowflake data warehouse. You need to select or configure your Snowflake destination.
 - d. Define the schedule for data ingestion, for example, how frequently you want the pipeline to pull data from the source.
 - e. Optionally, configure any data transformation or preprocessing rules in Hevo before the data gets loaded into Snowflake.
 - f. Monitor the pipeline stages—from data injection, transformation, schema mapping (to handle schema changes), to the final load into Snowflake.

2. Source Configuration:

- Type: PostgreSQL
- **Ingestion Mode: Logical Replication**
- further set `wal_level = logical`

3. Destination Configuration:

- Type: Snowflake Partner Connect
- Schema Name: `PC_HEVODATA_DB`

Example:

The image displays two screenshots of the Hevo dashboard interface. The top screenshot shows a pipeline configuration for a PostgreSQL Source connected to a Snowflake Partner Connect destination. The pipeline is active, and the status bar indicates 'Connected with Source', 'Started Ingesting Events', 'Staging Events for Loading', and 'Started Loading Events'. Below the status bar, the 'Pipeline Activity' section shows a progress bar for 'Logical Replication' at 33% completion, with 1/3 objects ingested. A table lists the objects being ingested: 'customer', 'feedback', and 'order', each with a 'Historical Load Ingested' status and a 'Position' value.

The bottom screenshot shows the same pipeline configuration, but the status bar indicates 'Connecting with Source' and 'Waiting for Events'. The 'Pipeline Activity' section shows a progress bar for 'Logical Replication' at 0% completion, with 0/3 objects ingested. The table lists the objects being ingested: 'customer', 'feedback', and 'order', each with a 'Historical Load Ingested' status and a 'Position' value.

2.4. Transformation Details (Objective 6)

2.4.1. Transformation 1:

- **Goal:** To create a new table in Snowflake as `order_events` where each row shows as `event_type = 'order_delivered'`.
- **Method Used:** Python
- **Reference:** <https://docs.hevodata.com/pipelines/transformations/python-transfm/utils/retrieve-the-type-for-value/>

- **Code:**

- Example

```
from io.hevo.api import Event
from io.hevo.api import Utils

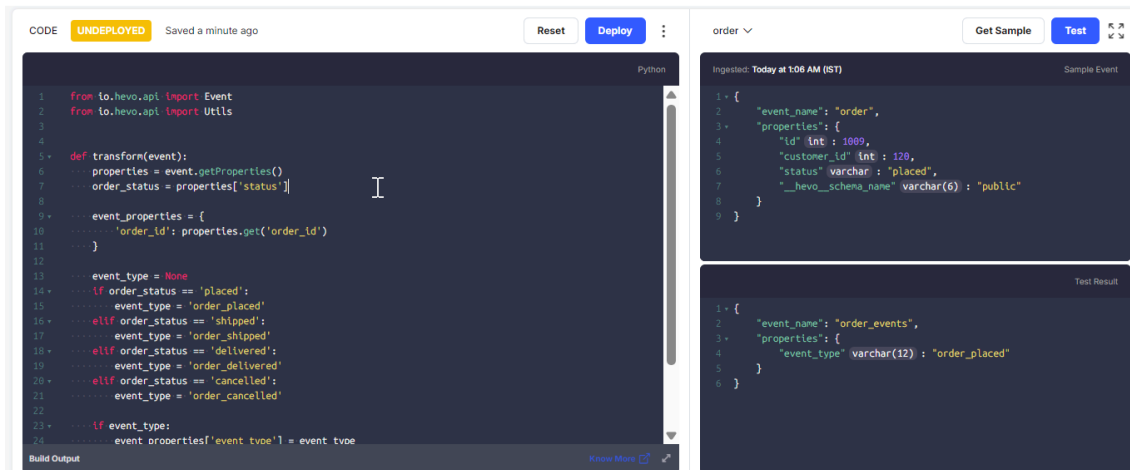
def transform(event):
    properties = event.getProperties()
    order_status = properties['status']

    event_properties = {
        'order_id': properties.get('order_id')
    }

    event_type = None
    if order_status == 'placed':
        event_type = 'order_placed'
    elif order_status == 'shipped':
        event_type = 'order_shipped'
    elif order_status == 'delivered':
        event_type = 'order_delivered'
    elif order_status == 'cancelled':
        event_type = 'order_cancelled'

    if event_type:
        event_properties['event_type'] = event_type
        new_event = Event('order_events', event_properties)
        return [new_event]

    return event
```



2.4.2. Transformation 2:

- **Goal:** Extract the text before the @ symbol from the email field as username
- **Method Used:** Python
- **Code:**
 - Sample

```
from io.hevo.api import Event
from io.hevo.api import Utils

def transform(event):
    properties = event.getProperties()

    email_field = properties.get('email')
    if isinstance(email_field, str) and '@' in email_field:
        username = email_field.split('@', 1)[0]
        properties['username'] = username
        event.setProperties(properties)
    return event
```

- **Example:** `jane.doe@gmail.com` creates a table with username `jane.doe`

2.5. Assumptions Made & Choices

- **status** **Field:** *Assumption:* The **status** field in the Postgres DDL was created as a **VARCHAR** instead of a custom Postgres **ENUM** for simplicity/wider compatibility.
 - **Transformation Language:** Python
-

2.6. Issues, Roadblocks, & Workarounds

- The main issue **revolved around** setting up the postgres set up I'm connecting the local database to the Hevo pipeline.
- The local database was not able to connect to the Hevo pipeline and to troubleshoot this we restarted the container restarted the service configured necessary as per the support article but still there were few network issues
- Using psql found out where are the necessary file to edit as prerequisite. Configured files **postgresql.conf** and **pg_hba.conf** by searching via psql **SHOW hba_file;** and **SHOW config_file;**
- Few times the database was able to connect to the servers using ngrok but still either due to high latency or some kind of network issues the further connection was not established
 - Multiple network issue addressed during the testing

ALERTS

PIPELINE #7
5 hours ago

PostgreSQL Source → Snowflake Partner Connect

Unable to connect to Source PostgreSQL Source

Failure Reason:

Unable to connect. Check the database host and port information. Ensure that the database host is accessible over the internet. If you are using a VPN or an on-premise database, whitelist Hevo's IP addresses.

Failed At:

Oct 17, 2025 21:50:14 IST

PIPELINE #7
5 hours ago

PostgreSQL Source → Snowflake Partner Connect

Failed to run

Failure Reason:

Unable to connect. Check the database host and port information. Ensure that the database host is accessible over the internet. If you are using a VPN or an on-premise database, whitelist Hevo's IP addresses.

Failed At:

Oct 17, 2025 21:49:23 IST

PIPELINE #7
5 hours ago

PostgreSQL Source → Snowflake Partner Connect

Source PostgreSQL Source is now reachable

Recovered At:

Oct 17, 2025 21:47:31 IST

PIPELINE #7
a day ago

PostgreSQL Source → Snowflake Partner Connect

Unable to connect to Source PostgreSQL Source

Failure Reason:

Unable to connect. Check the database host and port information. Ensure that the database host is accessible over the internet. If you are using a VPN or an on-premise database, whitelist Hevo's IP addresses.

Failed At:

Oct 17, 2025 01:50:04 IST

PIPELINE #7
a day ago

PostgreSQL Source → Snowflake Partner Connect

Your Pipeline Has Started Loading Data

Message:

Hi Siddharth, Your pipeline #7 has started loading data from the Source to your Destination. Please navigate to the Pipeline Overview page to view the Pipeline progress.

PIPELINE #7
a day ago

PostgreSQL Source → Snowflake Partner Connect

Processed the Pipeline again.

Recovered At:

Oct 17, 2025 01:06:21 IST

2.7. Validation Check on Snowflake:

The screenshot shows the Snowflake Database Explorer on the left and the Table Details for PC_HEVODATA_DB / PUBLIC / ENDTABLE_CUSTOMER on the right. The Database Explorer shows a tree view with PC_HEVODATA_DB selected, containing INFORMATION_SCHEMA, PUBLIC, SNOWFLAKE, SNOWFLAKE_LEARNING_DB, SNOWFLAKE_SAMPLE_DATA, and USER\$MASTER. The PUBLIC schema contains three tables: ENDTABLE_CUSTOMER, ENDTABLE_FEEDBACK, and ENDTABLE_ORDER. The Table Details view shows the table's metadata, including its location (PC_HEVODATA_ROLE), size (16.0KB), and a data preview. The data preview shows 100 of 200 rows, with columns ID, FIRST_NAME, LAST_NAME, EMAIL, and ADDRESS. The data is as of 1 day ago.

ID	FIRST_NAME	LAST_NAME	EMAIL	ADDRESS
1	Melissa	Elliott	anna31@ryan-brooks.com	{ "city": "Coxtown", "g
2	Marcus	Jefferson	raymond20@ayers.com	{ "city": "West Johnshi
3	Susan	Morrison	annaburton@collins.com	{ "city": "North Emily",
4	Ray	Johnson	garciafrank@barton-nelson.info	{ "city": "New Victorshi
5	Peter	Jordan	dmiller@yahoo.com	{ "city": "Lake Sandra",
6	Michael	Roth	gpineda@hotmail.com	{ "city": "East Dillonfort
7	Amanda	Lewis	barbara40@wilson.com	{ "city": "Ballhaven", "
8	Barbara	Green	thomasstephanie@hotmail.com	{ "city": "Port Shelbybu
9	Daniel	Reed	melissabrown@hotmail.com	{ "city": "Hillfurt", "sta
10	Shawn	King	victor01@chavez-rice.info	{ "city": "Smithton", "s
11	Doris	Sandoval	pgreen@yahoo.com	{ "city": "Port Lauren",
12	David	Robertson	clarkvictoria@gmail.com	{ "city": "Danielview",
13	Kelly	Barber	woodsusan@smith.net	{ "city": "Maryville", "s
14	Michelle	Stone	erinellison@gmail.com	{ "city": "West Aaronm

SQL validation on snowflake:

Sample: To get rows in table

```
SELECT
  'ENDTABLE_CUSTOMER' AS table_name,
  COUNT(*) AS row_count
FROM
  ENDTABLE_CUSTOMER;
```

The screenshot shows the Snowflake SQL Editor with the SQL query from the previous block. The query is executed, and the results are displayed in a table. The results show one row for the table ENDTABLE_CUSTOMER, with a row count of 200.

TABLE_NAME	ROW_COUNT
ENDTABLE_CUSTOMER	200

Sample: To get columns in table

```
SELECT
    TABLE_NAME,
    COUNT(COLUMN_NAME) AS column_count
FROM
    INFORMATION_SCHEMA.COLUMNS
WHERE

    TABLE_NAME = 'ENDTABLE_CUSTOMER'
GROUP BY
    TABLE_NAME
ORDER BY
    TABLE_NAME;
```

Replace table name with ENDTABLE_FEEDBACK, ENDTABLE_ORDER to get output for remaining tables.